OS UNIT4 ASSIGNMENT

PHASE 1:

Screenshot 1A — capture ./test_objects output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

Screenshot 1B — capture the sharded directory listing

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % find .pes/objects -type f

.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

PHASE 2:

Screenshot 2A — capture ./test_tree output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./test_tree

Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

Screenshot 2B — capture the xxd hex dump of the tree object

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "hello" > file1.txt
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes add file1.txt
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes commit -m "test"
Committed: 3e52b2a3ead0... test
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % find .pes/objects -type f | while read f; do if head -c 4 "
$f" | grep -q "^tree"; then echo "$f"; xxd "$f" | head -20; break; fi; done
.pes/objects/09/b576c2eb51b78fcf9b8493a0909522a75f8ee86df0e926312deeb5fbe7dfb1
00000000: 7472 6565 2034 3900 3130 3036 3434 2066  tree 49.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b4                   c.%.Wm...
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

PHASE 3:

Screenshot 3A — capture the full sequence output with status

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes init
  Initialized empty PES repository in .pes/
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "hello" > file1.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "world" > file2.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes add file1.txt file2.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes status
  Staged changes:
    staged:     file1.txt
    staged:     file2.txt

  Unstaged changes:
    (nothing to show)

  Untracked files:
    untracked:  index.h
    untracked:  object.c
    untracked:  pes.c
    untracked:  Makefile
    untracked:  test_objects
    untracked:  tree.h
    untracked:  test_objects.c
    untracked:  commit.c
    untracked:  test_tree
    untracked:  README.md
    untracked:  index.c
    untracked:  test_sequence.sh
    untracked:  .gitignore
    untracked:  test_tree.c
    untracked:  compile_flags.txt
    untracked:  pes.h
    untracked:  commit.h
    untracked:  tree.c
```

Screenshot 3B — capture the index file contents

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % cat .pes/index
  100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776656209 6 file1.txt
  100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776656214 6 file2.txt
○ bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

PHASE 4:

Screenshot 4A — capture ./pes log showing all 3 commits

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "Hello" > hello.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes add hello.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes commit -m "Initial commit"
  Committed: 0c2a3a9cfd55... Initial commit
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "World" >> hello.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes add hello.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes commit -m "Add world"
  Committed: 360873a8283c... Add world
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % echo "Goodbye" > bye.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes add bye.txt
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes commit -m "Add farewell"
  Committed: 5d46a66ad6fe... Add farewell
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % ./pes log
  commit 5d46a66ad6febf13d87c9b09a0ee44b971e7cb7070208cb741b0f612bd7dfa5d
  Author: PES User <pes@localhost>
  Date:   1776656579

      Add farewell

  commit 360873a8283cce525d12052228785526a7ddff219e39d8bdf759979837143096
  Author: PES User <pes@localhost>
  Date:   1776656560

      Add world

  commit 0c2a3a9cfd55cc76a77e5e602d5150c89be6873e622287eebc732b5463ea208a
  Author: PES User <pes@localhost>
  Date:   1776656540

      Initial commit

○ bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

Screenshot 4B — capture the full object store listing

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % find .pes -type f | sort

.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0c/2a3a9cfd55cc76a77e5e602d5150c89be6873e622287eebc732b5463ea208a
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/36/0873a8283cce525d12052228785526a7ddff219e39d8bdf759979837143096
.pes/objects/5d/46a66ad6febf13d87c9b09a0ee44b971e7cb7070208cb741b0f612bd7dfa5d
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

Screenshot 4C — capture both outputs

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % cat .pes/refs/heads/main
5d46a66ad6febf13d87c9b09a0ee44b971e7cb7070208cb741b0f612bd7dfa5d
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % cat .pes/HEAD
ref: refs/heads/main
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```

Final Screenshot:

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs % make test-integration

gcc -Wall -Wextra -O2 -I/opt/homebrew/opt/openssl@3/include -c object.c -o object.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 8e9c6349df3e... Initial commit

Log after first commit:
commit 8e9c6349df3ee97584b500e6672982445d0003d38b7e853e9f1e4c428c9b441f
Author: PES User <pes@localhost>
Date:   1776658835

    Initial commit


--- Second Commit ---
Committed: 659da0cc19b5... Update file.txt

--- Third Commit ---
Committed: 6cc309933376... Add farewell

--- Full History ---
commit 6cc309933376c5d23fb98fca8a81e853a330814ea897c140d817f2be72aa5537
Author: PES User <pes@localhost>
Date:   1776658835

    Add farewell

commit 659da0cc19b5beeb6f93815bb486ae9b982cce5e44a225044e011a90d38e2b53
Author: PES User <pes@localhost>
Date:   1776658835

    Update file.txt

commit 8e9c6349df3ee97584b500e6672982445d0003d38b7e853e9f1e4c428c9b441f
Author: PES User <pes@localhost>
Date:   1776658835

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6cc309933376c5d23fb98fca8a81e853a330814ea897c140d817f2be72aa5537

--- Object Store ---
Objects created:
      10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/65/9da0cc19b5beeb6f93815bb486ae9b982cce5e44a225044e011a90d38e2b53
.pes/objects/6c/c309933376c5d23fb98fca8a81e853a330814ea897c140d817f2be72aa5537
.pes/objects/8e/9c6349df3ee97584b500e6672982445d0003d38b7e853e9f1e4c428c9b441f
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS115-pes-vcs %
```